

1. Create new storage account

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information for 'tredence17@michaelklo...'. The main content area displays the 'Overview' page for a deployment named 'shibang_1756096946039'. A green checkmark indicates 'Your deployment is complete'. Key details include: Deployment name: shibang_17560969..., Subscription: LabsKraft, Resource group: LabsKraft2027, Start time: 8/25/2025, 10:12:53 AM, and Correlation ID: b4da9519-638e-45b6-9d58-b7b33b80ab37. A 'Go to resource' button is visible. The left sidebar shows navigation options: Overview, Inputs, Outputs, and Template. The right sidebar contains links for Cost Management, Microsoft Defender for Cloud, Free Microsoft tutorials, and Work with an expert.

2. Create a new container

The screenshot shows the Microsoft Azure portal interface with the 'New container' dialog box open. The dialog box has a 'Name' field with the value 'raw' and a green checkmark. The 'Anonymous access level' is set to 'Private (no anonymous access)'. A message states: 'The access level is set to private because anonymous access is disabled on this storage account.' The 'Advanced' section is collapsed. The 'Create' button is at the bottom left of the dialog. The background shows the 'Containers' page for the storage account 'shibang', with a list of containers (Name, \$logs) and a 'Last modified' column. The left sidebar shows navigation options: Overview, Activity log, Tags, Diagnose and solve problems, Access Control (IAM), Data migration, Events, Storage browser, Partner solutions, Resource visualizer, Data storage, Containers, File shares, Queues, and Tables.

3. Upload blob in raw container

The screenshot shows the Microsoft Azure portal interface. On the left, the navigation pane shows the 'raw' container under the 'shibang' storage account. The main area displays the 'raw' container details, including the authentication method (Access key) and a search bar for blobs. The 'Upload blob' dialog is open on the right, showing a file named 'retail_iot_data.csv' selected for upload. The dialog includes options to overwrite existing files and an 'Upload' button.

4. Create Access Connector -> name – shibang

The screenshot shows the Microsoft Azure portal interface. The main area displays the 'AccessConnectorCreate-20250825102024' deployment overview. The deployment is complete, and the 'shibang' resource is listed as 'Access Connector for Azure Data Lake Storage Gen2' with a status of 'Created'. The deployment details section shows the resource name, type, status, and operation. The 'Next steps' section includes a 'Go to resource' button.

Resource	Type	Status	Operation
shibang	Access Connector for Azure Data Lake Storage Gen2	Created	Operatic

5. Give role assignment

The screenshot shows the 'Add role assignment' page in the Microsoft Azure portal. The breadcrumb navigation is 'Home > shibang | Access Control (IAM) >'. The page title is 'Add role assignment'. There are tabs for 'Role', 'Members', 'Conditions', and 'Review + assign'. A message states: 'Showing a filtered list of roles because your permissions include a condition. [Learn more](#) [View my access](#)'. The 'Selected role' is 'Storage Blob Data Contributor'. Under 'Assign access to', 'Managed identity' is selected. The 'Members' section has a '+ Select members' link. Below is a table with columns 'Name', 'Object ID', and 'Type'. The table contains one entry: 'shibang' with Object ID '3689b3b5-b618-4e38-a3a7-03d0b7de5...' and Type 'Access Connector for Azure Databricks'. There is a 'Description' field with the text 'Optional'. At the bottom, there are buttons for 'Review + assign', 'Previous', and 'Next'. A feedback icon is in the bottom right corner.

Home > shibang | Access Control (IAM) >

Add role assignment

Role Members Conditions Review + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#) [View my access](#)

Selected role Storage Blob Data Contributor

Assign access to ☐ User, group, or service principal ☒ Managed identity

Members + Select members

Name	Object ID	Type
shibang	3689b3b5-b618-4e38-a3a7-03d0b7de5...	Access Connector for Azure Databricks

Description Optional

[Review + assign](#) [Previous](#) [Next](#) [Feedback](#)

6. Create Synapse workspace

The screenshot shows the 'shibang' Synapse workspace overview page in the Microsoft Azure portal. The breadcrumb navigation is 'Home >'. The page title is 'shibang Synapse workspace'. There is a search bar and a 'Suggest a workload for this Synapse workspace' button. The left sidebar has a search bar and a list of navigation items: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Resource visualizer, Settings, Analytics pools, Security, Monitoring, Automation, and Help. The main content area has a top bar with '+ New dedicated SQL pool', '+ New Apache Spark pool', 'Refresh', 'Reset SQL admin password', and 'Delete'. Below this is the 'Essentials' section with a 'JSON View' link. The 'Essentials' section contains the following information: Resource group (move) [LabsKraft2027](#), Status Succeeded, Location Poland Central, Subscription (move) [LabsKraft](#), Subscription ID d008c7cc-9795-4292-bf79-74ae52cda63d, Managed virtual network No, Managed Identity object ID 6f3489b4-5ac8-4a7d-ae02-435eb8359090, Workspace web URL <https://web.azuresynapse.net?workspace=%2fsubscriptions%2fd008c7cc-...>, Tags (edit) [Add tags](#), Networking [Show firewall settings](#), Primary ADLS Gen2 account URL <https://shibang.dfs.core.windows.net>, Primary ADLS Gen2 file system raw, SQL admin username sqladminuser, SQL Microsoft Entra admin tredence17@michaelkloudkraft@hotmail.onmicrosoft.com, Dedicated SQL endpoint shibang.sql.azuresynapse.net, Serverless SQL endpoint shibang-ondemand.sql.azuresynapse.net, Development endpoint <https://shibang.dev.azuresynapse.net>. At the bottom, there is a 'Getting started' section.

Click to go back, hold to see history

Home >

shibang Synapse workspace

Suggest a workload for this Synapse workspace

Search

+ New dedicated SQL pool + New Apache Spark pool Refresh Reset SQL admin password Delete

Overview [JSON View](#)

Essentials

Resource group (move) [LabsKraft2027](#)

Status Succeeded

Location Poland Central

Subscription (move) [LabsKraft](#)

Subscription ID d008c7cc-9795-4292-bf79-74ae52cda63d

Managed virtual network No

Managed Identity object ID 6f3489b4-5ac8-4a7d-ae02-435eb8359090

Workspace web URL <https://web.azuresynapse.net?workspace=%2fsubscriptions%2fd008c7cc-...>

Tags (edit) [Add tags](#)

Networking [Show firewall settings](#)

Primary ADLS Gen2 account URL <https://shibang.dfs.core.windows.net>

Primary ADLS Gen2 file system raw

SQL admin username sqladminuser

SQL Microsoft Entra admin tredence17@michaelkloudkraft@hotmail.onmicrosoft.com

Dedicated SQL endpoint shibang.sql.azuresynapse.net

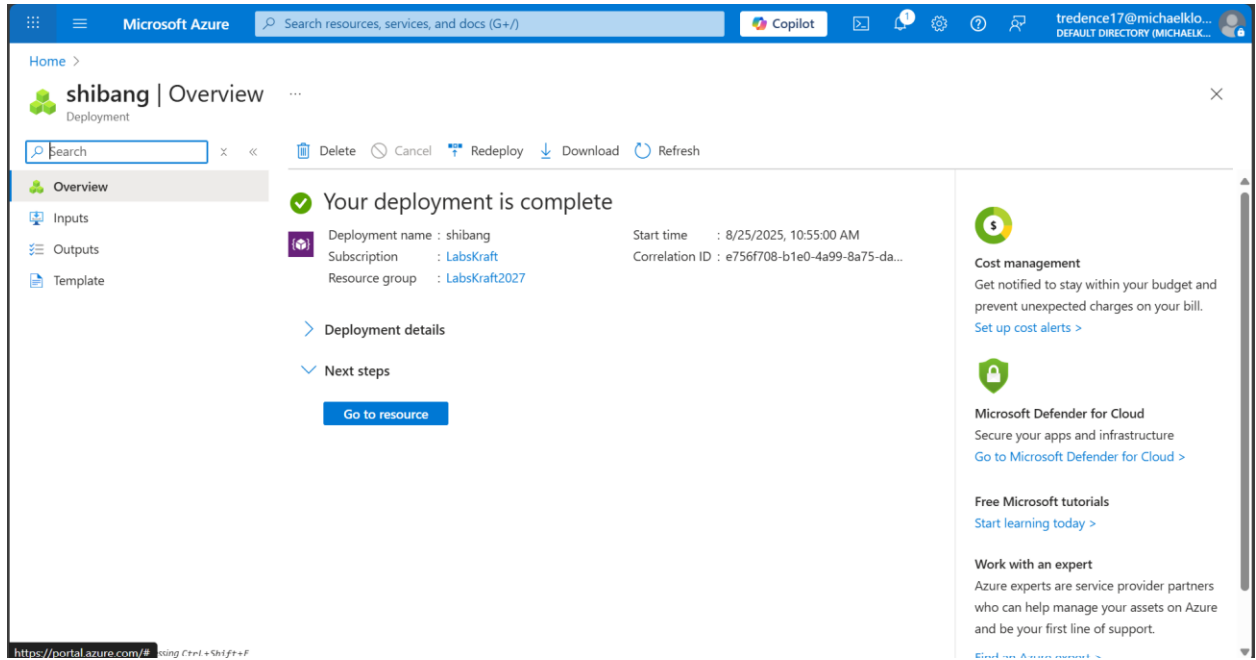
Serverless SQL endpoint shibang-ondemand.sql.azuresynapse.net

Development endpoint <https://shibang.dev.azuresynapse.net>

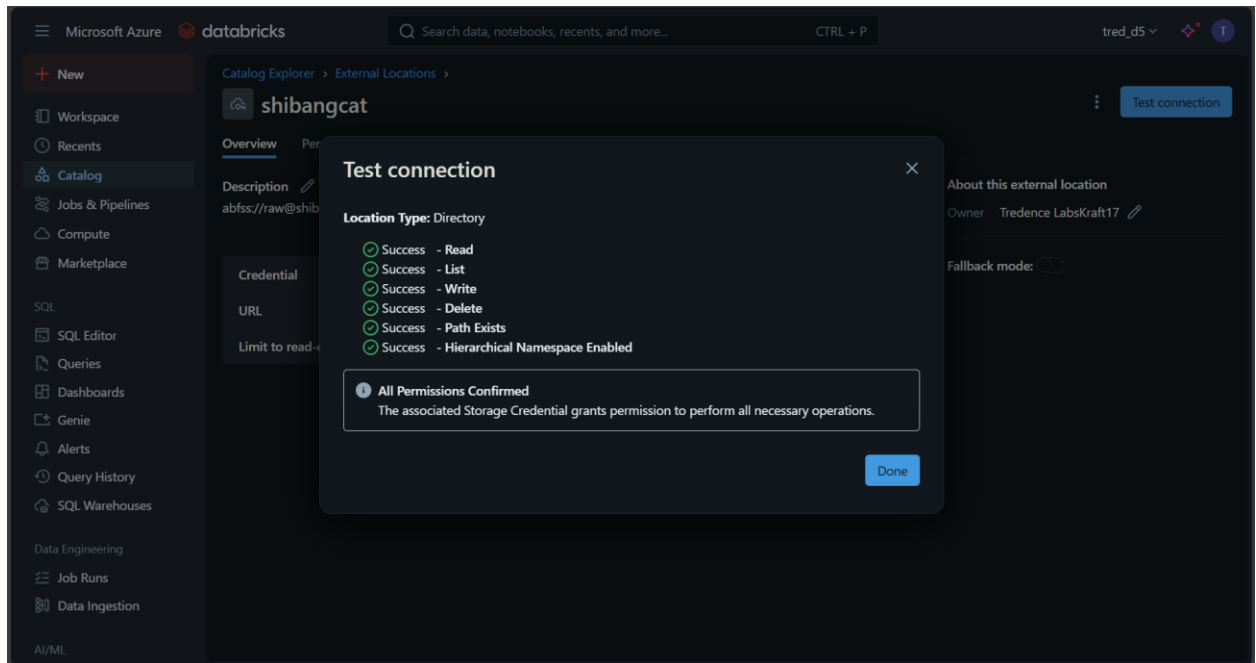
Getting started

Add or remove favorites by pressing Ctrl+Shift+F

7. Create dedicated pool
8. Create event hub



9. Create a new catalog's external location



10. Create new catalog

The screenshot shows the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar is open, displaying a tree view of catalogs under 'My organization'. The 'shibang' catalog is selected. The main panel shows the 'shibang' catalog details, including a search bar for schemas, a table of schemas, and a section for 'About this catalog'.

Name	Owner	Created at
default	tredence17@michaelkloudkraft...	Aug 25, 2025, 11:11 AM
information_schema	System user	Aug 25, 2025, 11:11 AM

About this catalog
Owner: Tredence LabsKraft17

11. Create bronze, silver and gold schema

The screenshot shows the Databricks Catalog Explorer interface, now displaying the 'bronze' schema within the 'shibang' catalog. The 'Catalog Explorer' sidebar shows the 'bronze' schema selected. The main panel shows the 'bronze' schema details, including a search bar for tables, a table of tables (which is empty), and a section for 'About this schema'.

About this schema
Owner: Tredence LabsKraft17

You either don't have access to any tables or there are no tables in this schema. Either contact your administrator or [create a new sample table](#).

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

tred_d5

New

Workspace

Recents

Catalog

Jobs & Pipelines

Compute

Marketplace

SQL

SQL Editor

Queries

Dashboards

Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

Catalog

Starter Warehouse Pro S

Type to search...

My organization

tred_d5

system

shibang

bronze

default

information_schema

silver

shiv_catalog

cap1_shiv_catalog

shiv

shivam_catalog_new

shivam-catalog

shivam-catalog-project

st_capstone_retail

st-catalog

Delta Shares Received

samples

Legacy

Catalog Explorer shibang

silver

Use with BI tools

Share

Create

Overview Details Permissions

Description

AI generate Add

Filter tables

Tables 0

Volumes 0

Models 0

Functions 0

Sort

Name Owner Created at Popularity

You either don't have access to any tables or there are no tables in this schema. Either contact your administrator or create a new sample table.

About this schema

Owner Tredence LabsKraft17

Tags

Add tags

Microsoft Azure databricks

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tred_d5

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Data Ingestion

AI/ML

Catalog

Starter Warehouse Pro S

Type to search...

My organization

tred_d5

system

shibang

bronze

default

gold

No data

information_schema

silver

shiv_catalog

cap1_shiv_catalog

shiv

shivam_catalog_new

shivam-catalog

shivam-catalog-project

st_capstone_retail

st-catalog

Delta Shares Received

Catalog Explorer shibang

gold

Use with BI tools

Share

Create

Overview Details Permissions

Description

AI generate Add

Filter tables

Tables 0

Volumes 0

Models 0

Functions 0

Sort

Name Owner Created at Popularity

You either don't have access to any tables or there are no tables in this schema. Either contact your administrator or create a new sample table.

About this schema

Owner Tredence LabsKraft17

Tags

Add tags

12. Create application

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

tredence17@michaelklo...
DEFAULT DIRECTORY (MICHAELKLO...)

Home > App registrations > shibang

Search

Overview

Quickstart

Integration assistant

Diagnose and solve problems

Manage

Branding & properties

Authentication (Preview)

Certificates & secrets

Token configuration

API permissions

Expose an API

App roles

Owners

Roles and administrators

Manifest

Delete Endpoints Preview features

Got a second? We would love your feedback on Microsoft identity platform (previously Azure AD for developer). →

Essentials

Display name
shibang

Application (client) ID
1a7520b5-735c-4235-91f1-6f89078920ee

Object ID
4d9f486f-077e-4a93-a04b-a52b3dfded64

Directory (tenant) ID
5950e39e-81d1-45a1-8618-fe39a39b0448

Supported account types
[My organization only](#)

Client credentials
[Add a certificate or secret](#)

Redirect URIs
[Add a Redirect URI](#)

Application ID URI
[Add an Application ID URI](#)

Managed application in local directory
shibang

Welcome to the new and improved App registrations. Looking to learn how it's changed from App registrations (Legacy)? [Learn more](#)

Starting June 30th, 2020 we will no longer add any new features to Azure Active Directory Authentication Library (ADAL) and Azure Active Directory Graph. We will continue to provide technical support and security updates but we will no longer provide feature updates. Applications will need to be upgraded to Microsoft Authentication Library (MSAL) and Microsoft Graph. [Learn more](#)

Get Started Documentation

13. Create new notebook

Microsoft Azure databricks

Search data, notebooks, recents, and more... CTRL + P

tred_ds

New

Workspace

Recents

Catalog

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Genie

Alerts

Query History

SQL Warehouses

Data Engineering

Job Runs

Data Ingestion

AI/ML

shibang-caps-1

schema_shibang

File Edit View Run Help Python Tabs: ON ☆ Last edit was 4 minutes ago

Run all Connected Share

Last execution failed

```
1 from dlt import table
2 import pyspark.sql.functions as F
3 from pyspark.sql.types import StructType, StructField, StringType, DoubleType, IntegerType, TimestampType
4 # :white_check_mark: Define schema for retail IoT CSV
5 iot_schema = StructType([
6     StructField("OrderID", StringType(), True),
7     StructField("CustomerID", StringType(), True),
8     StructField("Region", StringType(), True),
9     StructField("StoreID", StringType(), True),
10    StructField("Product", StringType(), True),
11    StructField("Quantity", IntegerType(), True),
12    StructField("UnitPrice", DoubleType(), True),
13    StructField("SalesAmount", DoubleType(), True),
14    StructField("OrderTime", StringType(), True), # can cast to Timestamp later in Silver
15    StructField("DeviceType", StringType(), True),
16    StructField("Temperature", DoubleType(), True),
17    StructField("DeviceStatus", StringType(), True),
18    StructField("AnomalyFlag", IntegerType(), True),
19    StructField("AnomalyType", StringType(), True)
20 ])
21 # :white_check_mark: Paths (use subdirectories to avoid overlap with DLT system files)
22 raw_path = "abfss://raw@shibang.dfs.core.windows.net/raw_data/"
23 schema_path = "abfss://raw@shibang.dfs.core.windows.net/dlt_schemas/bronze_orders_shibang"
24 checkpoint_path = "abfss://raw@shibang.dfs.core.windows.net/dlt_checkpoints/bronze_orders_shibang"
```

14. Create new schema

15. Create etl pipeline

16. Pipeline with bronze, silver and gold layer

17. PowerBI dashboard

